Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/10/2016
PI Name:	Garofalo, Roberto M.D.		
Project Title:	Effect of Space Flight on Innate Immunity to Respiratory Viral Infections		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	77555-0366	**Congressional District:**	14
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-133 ; ISS NOTE: End date is now 9/30/2015 per NSSC information (Ed., 4/14/15) NOTE: Extended to 4/30/2015 per A. Chu/ARC and NSSC information (Ed., 4/3/2014) NOTE: Extended to 4/30/2014 per NSSC information (Ed., 1/31/13) NOTE: Extended to 4/30/2013 per A. Chu/ARC (Ed., 3/6/12)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Casola, Antonella (University of Texas Medical Branch)		
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Task Description:

Respiratory Syncytial Virus (RSV): Human respiratory syncytial virus (RSV) is considered the leading cause of lower respiratory tract infection in infants and children worldwide, and may cause severe and even fatal lung disease in immunocompromised and elderly persons. Approximately two-thirds of infants are infected with RSV during the first year of life, and 90% have been infected one or more times by 2 years of age. However, both infants and adults are susceptible to recurrent infections, and accumulating evidence suggests a strong link between RSV bronchiolitis and later development of asthma-like syndromes. Previous reports show that the intensity and duration of the immune response to RSV may be determined by the local cellular and cytokine response. To date, there is no safe and effective vaccine available, treatment options against RSV infection is limited.
The goal of the Mouse Immunology-2 experiment is to discover what triggers and leads to an increased susceptibility to an infection. The overarching hypothesis is that combinations of several factors, from microgravity to stress, affect the human immune system during space flight, resulting in an impaired antiviral response in the respiratory tract.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Space Applications: Space shuttle Discovery will bring NASA scientists one step closer to helping astronauts and the public discover ways to battle and prevent serious illness and infection. These investigations are expected to generate new scientific evidence of the immune pathways that are affected in antiviral host response during space flight and possible preventive or therapeutic approaches applicable to future space missions.
Earth Applications: Understanding the function of the immune system during space flight may have great relevance to our understanding of the process of aging and/or stress-related immune-modulation (adjustments in the level of an immune response) on Earth. In particular, studying the innate host response against pathogens during/after space flight will provide novel data on the function of the respiratory mucosal response to viral pathogens.

The goal of the Mouse Immunology-2 experiment is to discover what triggers and leads to an increased susceptibility to an infection. These findings can be used to help treat and prevent future astronauts from getting sick, as well as protect people with more vulnerable immune systems here on Earth, such as the elderly or young children.

Task Progress:

Specifically our experiment objectives are:
Aim 1: Effect of RSV infection and space flight on oxidative stress. To assess the oxidative responses in lung tissue to viral infection. These experiments will be conducted in mice flown in the Space Shuttle that will be infected with RSV immediately after return to Earth;

Aim 2: Space flight and RSV infection induces alterations of H2S pathways. To investigate the effect of the key H2S generating enzymes CSE (cystathionine gamma-lyase), CBS (cystathionine-beta synthase), and 3MST (3-mercaptopyruvate sulfurtransferase) during the course of infection in lung samples collected from FLT (flight), AEM (animal enclosure module), and VIV (vivarium) mice. These studies will be conducted in mice as in Aim1.

Effect of RSV infection and space flight on oxidative stress. Over the past ten years we discovered that in the course of respiratory viral infections, such as those caused by respiratory syncytial virus (RSV), reactive oxygen species (ROS) are rapidly generated but do not contribute to the host (innate) anti-viral responses, rather they cause lung inflammation and clinical disease. Oxidative stress plays an important role in the pathogenesis of lung inflammation. We have previously shown that RSV is a potent inducer of ROS in airway epithelial cells in vitro and that RSV causes significant oxidative stress lung damage in vivo, as demonstrated by the increase of lipid peroxidation markers such as F2 8-isoprostane, MDA, and 4-HNE in an animal model of infection. To further define the role of stress and space flight, future experiments were planned to assess the oxidative responses in lung tissue to viral infection. This data provide new insights in the design of countermeasures to tissue oxidative damage associated with space exploration. Several previous studies have shown the occurrence of oxidative stress during space flight. However our studies will give a first insight into the impact of space flight associated factors on oxidative stress damages to the lungs of BALB/c mice to viral infection. These experiments were conducted in lung tissue samples harvested from mice flown on the Space Shuttle Discovery, and compared with the ground controls groups. Levels of antioxidant enzymes (AOEs) superoxide dismutase (SOD), and catalase were measured in harvested lungs tissue from mice exposed to space flight environment, and compared with vivarium ground control groups. Similar, nuclear levels of NF-E2-related factor 2 (Nrf2), a transcription factor which upregulates basal and inducible expression of AOE genes, were measured in lung tissue. We demonstrate that space flight environment induces changes in antioxidant enzymes (AOEs) Nrf2, Catalase, and SOD levels in FLT RSV infected mice compared with ground control groups.

Space flight and RSV infection induces alterations of H2S pathways. Recently, our laboratory has made the critical discovery that changes in cellular H2S significantly affects proinflammatory cellular responses and viral replication both in vitro and in vivo models of RSV infection. We have characterized a novel critical component of the innate response to viral pathogens, which is regulated by the endogenous gasotransmitter hydrogen sulfide (H2S) and could link immune response to the oxidative pathway. It is currently unknown if environmental conditions that cause redox imbalance and inflammation in the host, such as those to which astronauts are exposed during space flights, affects endogenous H2S production and/or the overall oxidative response. To test the hypothesis that H2S modulates viral replication, clinical disease, and lung inflammation in response to viral infection, we investigated the fate of the key H2S generating enzymes CSE (cystathionine gamma-lyase), CBS (cystathionine-beta synthase), and 3MST (3-mercaptopyruvate sulfurtransferase) during the course of infection by investigating their expression in lung samples collected from FLT, AEM, and VIV mice. We found that CSE, CBS, 3-MST mRNA expression levels were significantly decreased by RSV infection . Overall, we provide evidence that FLT mice infected with RSV displayed a decreased ability to generate H2S-enzymes CSE, CBS, and 3-MST in lung tissue when compared with VIV RSV infected mice or sham inoculated ones.

In summary, our findings suggest that in a BALB/c mouse model of RSV infection, a relatively short 12.8-day flight in space induced significant changes in parameters that can influence immune defense mechanisms, changes that may become significant during long-term space missions. Differences between the groups (VIV, AEM, and FLT) in several immune parameters emphasize the importance of simulating the space flight conditions, when comparisons are made between animals in flight and ground controls. This first-time immune evaluation of the BALB/c mouse model to space flight environment and viral infection opens the possibility of expanded research which could provide further better understanding in the mechanisms responsible for the observed changes in the immune parameters in order to reduce health risks to astronauts.

	Further data analysis is in progress to characterize oxidative gene expression in RSV-infected mice that flew in space as well as in ground-based mouse models of space flight: We will test the hypothesis that hindlimb unloading (HLU) model alters the expression of antioxidant (AOEs) and H2S enzymes levels in a BALB/c mouse model of viral infection, and the results will be compared with space flight data from STS-133 mission. We will use NASA's hindlimb suspended model of exposure to microgravity to monitor the oxidative stress in the lung tissue of BALB/c mouse model of viral infection. Levels of antioxidant enzymes (AOEs) superoxide dismutase (SOD) and catalase will be measured in harvested lungs tissue following HLU, and will be compared with vivarium ground control groups. Similar, nuclear levels of NF-E2-related factor 2 (Nrf2), a transcription factor, will be measured in lung tissue. Moreover, we will investigate the fate of the key H2S generating enzymes CSE (cystathionine gamma-lyase), CBS (cystathionine-beta synthase), and 3MST (3-mercaptopyruvate sulfurtransferase) during the course of infection in a HLU mouse model of infection. Our goal is to help bridge the knowledge gap in terms of pulmonary complications from space travel through the development and testing of a novel hindlimb unloading in vivo model of viral infection. PUBLICATIONS AND POSTERS SUPPORTED BY NASA GRANT: 1. Ivanciuc T., Casola A., Garofalo, RP. Oral presentation: Oxidative lung damage and the role of H2S on space flight environment in a mouse model of respiratory syncytial virus (RSV). NASA Human Research Program Investigators' Workshop Frontiers in Human Space Exploration Research, February 8-11, 2016, Galveston Island Convention Center, Galveston, TX 2. Garofalo, RP, Casola, A., Ivanciuc, T. Oral presentation: Redox-Sensitive Antiviral Immunity and Space Flight, ASGSR, Nov 11-14, 2015, Alexandria, VA. 3. Ivanciuc, T., Kolli, D., Dumars, P., Sbrana, E., Hosakote, Y., Blell, Z., Rayl, N., Casola, A., Garofalo, R.P. Effect of space flight on innate immunity to respiratory viral infections – manuscript in revision 4. Ivanciuc, T., Sbrana, E., Casola, A., & Garofalo, R.P. Hindlimb unloading in mice as ground-based model of space flight to study immunity to respiratory viral infections – manuscript in revision.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Abstracts for Journals and Proceedings	Ivanciuc T, Casola A, Garofalo RP. "Oxidative lung damage and the role of H2S on space flight environment in a mouse model of respiratory syncytial virus (RSV)." Oral presentation. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016
Abstracts for Journals and Proceedings	Garofalo RP, Casola A, Ivanciuc T. "Redox-Sensitive Antiviral Immunity and Space Flight." Oral presentation. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015